Instructions

Close ×

Your Task

In this lab, you will implement a distributed array summation system using MPI (Message Passing Interface) with scatter and gather operations. This assignment will help you understand the concepts of collective communication in distributed computing and practice using `MPI_Scatter` and `MPI_Gather`.

Background

The provided C++ program divides a large array among multiple processes. Each process computes the partial sum of its segment, and then these partial sums are collected and combined to compute the total sum.

Implementation Tasks

1. Complete the TODO sections in the code to implement the `MPI_Scatter` and `MPI_Gather` operations.
2. Compile and run the program using multiple MPI processes (e.g., 4 or 8).
3. Analyze the output to ensure:
    a. The array is correctly distributed among processes.
    b. Each process computes the correct partial sum.
    c. The final sum is correctly calculated and matches the expected value.

Required MPI Calls

- `MPI_Scatter`: Used to distribute parts of the main array from the root process to all participating processes.
- `MPI_Gather`: Used to collect partial sums from each process back to the root process.

To Compile and Run

Submission Instructions

- `array_sum.cpp`: Your completed C++ source file with working `MPI_Scatter` and `MPI_Gather` calls.

Mark as completed